

Perspective

Recent Trends in Quantum Chemical Modeling of Enzymatic Reactions

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Recent Trends in Quantum Chemical Modeling of Enzymatic Reactions

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Abstract

The quantum chemical cluster approach is a powerful method for investigating enzymatic reactions. Over the last two decades, a large number of highly diverse systems have been studied and a great wealth of mechanistic insight has been developed using this technique. This Perspective reviews the current status of the methodology. The latest technical developments are highlighted and challenges are discussed. Some recent applications are presented to illustrate the capabilities and progress of this approach, and likely future directions are outlined.

1. Introduction

The use of limited models to study the active sites of enzymes, the so-called quantum chemical cluster or all-QM approach, has over the past twenty years gradually been refined and is today a very important tool for the elucidation of reaction mechanisms and other properties of enzymes.[1] The basic idea of this methodology is to focus on a small part of the enzyme around the active site and treat it with relatively accurate quantum chemical methods. Density functional theory (DFT) has been the electronic structure method of choice in the cluster approach. In particular, the B3LYP functional [2] has been used extensively over the years as it has been found to provide a good balance between speed and accuracy. Many of the ideas and computational techniques used in the cluster approach bear high similarities to those used in the field of computational homogeneous catalysis, in which the quantum chemical methodology has been extremely successful and has contributed enormously to the understanding of catalytic processes and the development of new experimental protocols.[3]

15-20 years ago, a typical active site model consisted of less than 50 atoms, and it was, somewhat surprisingly, possible with the cluster approach to address and solve some mechanistic problems, in particular concerned with metalloenzymes.[4] As computers over time have become faster and cheaper, the active site models have accordingly gradually become larger, and models consisting of 250-300 atoms can today be considered as routine. The cluster approach has developed into a robust scheme and has proven to be highly versatile. It has been applied to a wide variety of enzyme families and, indeed, a large number of complex problems have been solved over the years.[5]

There have been a number of reviews describing the cluster approach and its utilization in different classes of enzymes.[1] In this Perspective, we will review some of the recent progress in this methodology and its applications, and discuss how the field is likely to develop over the coming years. First, some technical details will be described and some recent trends to improve the approach will be reviewed. Next, a number of very recent examples will be presented, mainly from our own work, to illustrate some of the current trends in the applications. For space reasons, only limited discussions regarding the experimental backgrounds of the discussed enzymes will be provided here.

Note that the cluster approach is widely used to model spectroscopic properties of enzyme active sites.[6] These applications will not be discussed here, as focus will be on modeling enzymatic reactions. Also, the important quantum mechanics/molecular mechanics (QM/MM) and empirical valence bond (EVB) methods for studying enzyme catalysis will not be discussed at all in the present Perspective, as they constitute entirely different approaches. There are a number of excellent recent reviews that can be consulted on these techniques.[7,8]

2. Methodological Developments

In the cluster approach, only a small part of the enzyme is included in the model to study the properties and reaction mechanism. The effects of the rest of the enzyme, the part not included in the model, is usually accounted for by two simple approximations. The steric influence that the enzyme matrix imposes on the active site is modeled by a coordinate locking scheme, in which a number of atoms are fixed to their crystallographic positions, typically where truncation is made. If the model is too small, this procedure could lead to artificial strain that results in wrong energy profiles. Larger models usually grant enough flexibility to accommodate changes that take place during the reaction and these models have been shown to yield generally good results. When one starts to approach models of more than 300 atoms, however, one issue that arises is the multiple minima problem, and one has to be very careful to assure that no artificial movements take place between the various stationary points, which could lead to wrong energy profiles and ultimately wrong mechanistic conclusions. This has in recent applications been solved by simply fixing more atoms around the truncation points, typically the carbon atom and one or two of the hydrogens that replace the connecting atoms.[1b] In general, it is always a good idea to start the mechanistic investigations with a relatively small model and then increase the size. This way one can examine the stability of the results, detect possible multiple minima errors in the large model, and also gain chemical insight into the roles of the various groups.

One possible way to replace the atom fixation scheme could be to introduce energy potentials that would allow the atoms to move at some energy penalty. To our knowledge, this has not been used in the context of the cluster approach and could possibly improve the

approximation.

The other approximation in the cluster approach is the use of implicit solvation models to account for the electrostatic influence of the enzyme surrounding. This method assumes that the enzyme surrounding is a homogenous polarizable medium, with a dielectric constant ϵ . In most applications, ϵ has been set to equal 4, which is somewhat arbitrary. However, a number of systematic studies have demonstrated that the solvation effect saturates quite rapidly with the size of the model [9], and the approximation is therefore expected to work better with increasing model size. Here, it is important to stress that if the obtained energies in the cluster model are very sensitive to the choice of the dielectric constant, then conclusions involving these energies must be considered as weak.

Elucidation of enzyme reaction mechanisms with the cluster approach relies on the use of electronic structure methods that can treat relatively large systems relatively accurately. As mentioned above, DFT has been the overwhelmingly most used method in this context. The hybrid functional B3LYP [2] has been particularly dominating in the applications, as it has been considered to represent a good trade-off between accuracy and speed. A recent development in this respect has been the addition of an empirical dispersion correction to the B3LYP results to remedy the known weakness of this functional, and indeed most other functionals, in their lack of description of the attractive dispersion interaction.[10] The technique, called DFT-D, has been shown to yield a significant improvement of the energies in the field of homogeneous catalysis.[11] Also in the context of enzyme modeling it has been demonstrated to improve both energies and geometries [12] and has quickly become a standard choice in the cluster approach. Another way to account for dispersion is to use functionals that include weak interactions in the training set, such as the Minnesota M06 suite [13]. These methods are quite common in the field of homogeneous catalysis, and are starting to be used also in enzyme modeling.[14,15]

In this context, another interesting development that has become possible due to the increased computer power is the use of highly correlated *ab initio* methods to obtain more accurate energies in the cluster approach.[14,16] The study of the oxygen evolving center in photosystem II with the density matrix renormalization group (DMRG) method is an exciting case.[17] This kind of applications are likely to become more frequent in the future as the

computational power will allow the treatment of ever larger systems.

The calculated energies are usually compared to experimental rates by means of classical transition state theory, an approximation that works well for most applications. Tunneling effects are normally not considered in the cluster approach as these in general contribute quite little to the absolute barriers. However, in cases where one is interested in reproducing kinetic isotope effects for the purpose of establishing a certain reaction mechanism, tunneling effects can be included using more or less sophisticated approximations.[18]

Another technical point here concerns the treatment of entropy in the cluster approach. The approach focuses entirely on the chemical step of the enzymatic reaction and starts from the enzyme-substrate complex. That is, the substrate binding and product release events are not considered at all. In general, entropy effects are rather small in the chemical steps of enzymatic reactions [19,20] and it is therefore a rather good approximation to simply neglect the entropy and approximate the free energy with the enthalpy. However, in cases where a gas molecule enters or exits during the reactions, entropy effects become significant and have to be included somehow. One way of dealing with these cases has in a number of applications been to estimate the entropy to be equal to the translational entropy of the free gas molecule. This has yielded satisfactory results in cases involving, e.g., the binding of an NO molecule or the release of CO₂. [21,22,23]

From modeling point of view, a case more problematic than dealing with gas molecules entering or exiting the active site model is when protons or electrons do so, that is, to calculate pK_a values or redox potentials.[24] These situations are rather common in enzyme chemistry, and in many cases a proper modeling of these properties is crucial in order to understand the entire reaction mechanism of the enzyme. Since the total charge of the model changes, the calculated energies become very sensitive to the surrounding and therefore rather unreliable. One pragmatic way to deal with this issue in the cluster approach is to include redox potentials or pK_a values directly from experiments if available, like, for example, in the case of the hydrolysis reaction taking place at the ribosome (discussed below).[15]

A fruitful approach that has been applied for a number of redox enzymes is to consider the driving force for an entire catalytic cycle, which in many cases is available from experiments, and introduce one parameter such that this driving force is reproduced.[1a,1b,25] To obtain

individual pK_a values in the cycle, one more experimental value is needed. This procedure has been used successfully in very advanced applications, like, for example, the mechanisms of photosystem II and cytochrome c oxidase.[26,27]

3. Trends in applications

Unraveling the mechanism by which an enzyme performs its reaction requires the generation and evaluation of energy profiles for various possible pathways, which entails performing a large number of calculations of intermediates and transition states. The cluster methodology offers a robust and sufficiently accurate protocol to perform this job. It enables a comparatively fast examination of different mechanistic proposals and the discrimination between them on the basis of their calculated energies. Indeed, over the past two decades, a large number of mechanisms have been studied for a wide variety systems, including some of the most complicated metalloenzymes.[1] The elucidation of the mechanism of oxygen formation at the oxygen evolution center of photosystem II is an excellent example of the capabilities of the cluster approach.[26] In what follows, a few selected examples of recent applications of the cluster methodology will be briefly presented to illustrate the various concepts discussed above and to demonstrate the versatility of the approach.

3.1. Mechanism of peptidyl-tRNA hydrolysis in ribosome.

One interesting recent application concerns the mechanism of the termination of protein synthesis on the ribosome.[15] This process, the hydrolysis of an ester bond between the P-site tRNA and the nascent peptide chain, is known from experiments to be pH-dependent involving an ionizable group with a pK_a higher than 9.[28] A cluster model (224 atoms) of the peptidyl transferase center of the ribosome was thus designed as shown in Figure 1A to investigate the peptidyl-tRNA hydrolysis reaction mechanism. First, the mechanisms involving stepwise or concerted six- or eight-membered transition states, analogous to those proposed for the ester aminolysis of the peptide bond formation that takes place at the same peptidyl transferase center, were investigated and found to be associated with high energy barriers (>30 kcal/mol).

Instead, the calculations suggested that the reaction is initiated by the deprotonation of the P-site A76 2'-OH group, which then acts as a general base to activate the water nucleophile (Figure 1B).[15] As discussed above, absolute protonation/deprotonation energies are difficult to calculate accurately. Therefore, the energy of the first step, the release of the 2'-OH proton to bulk, was estimated from the difference between the experimental pK_a of a model compound (3'-O-methyladenosine; $pK_a = 13.7$) and the pH of the reaction ($pH = 7.5$), yielding an energy of $1.36 \times (13.7 - 7.5) = +8.4$ kcal/mol. Next, the barrier for the following concerted nucleophilic attack of the water and proton transfer to the 2'-O group is calculated to be only 7.4 kcal/mol, resulting in an overall activation energy of 15.8 kcal/mol. The subsequent collapse of the tetrahedral intermediate is calculated to be very fast, with a low barrier (Figure 1B). This proposed mechanism involving the ionization of A76-O2' is consistent with the pH dependence of the reaction, and the calculated overall barrier is consistent with the measured kinetics of the reaction. This kind of mechanistic studies represents one of the great successes and strengths of the cluster approach and it will undoubtedly continue in the future.

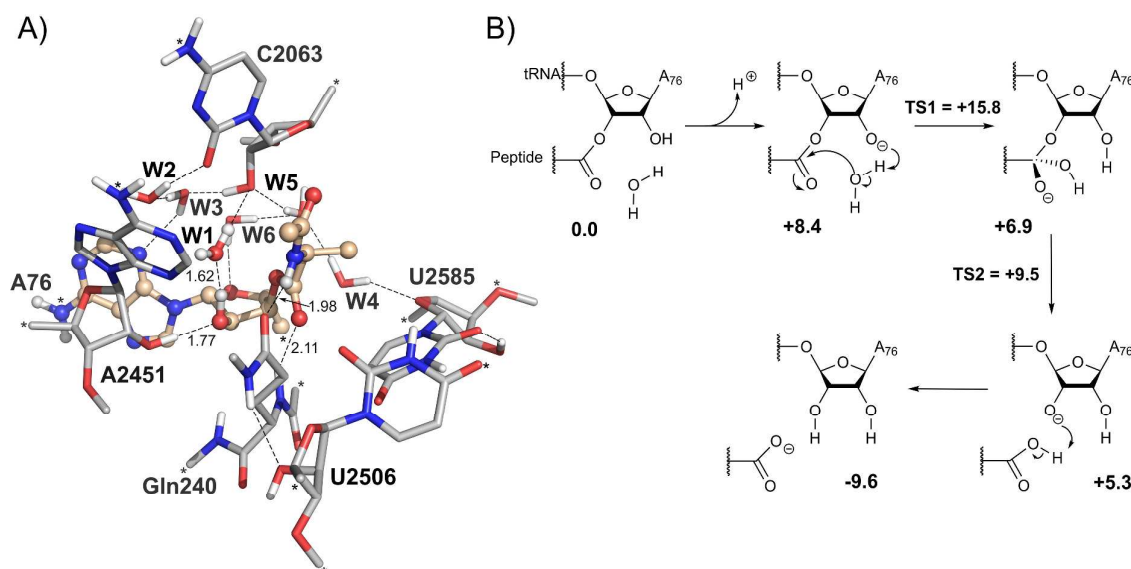
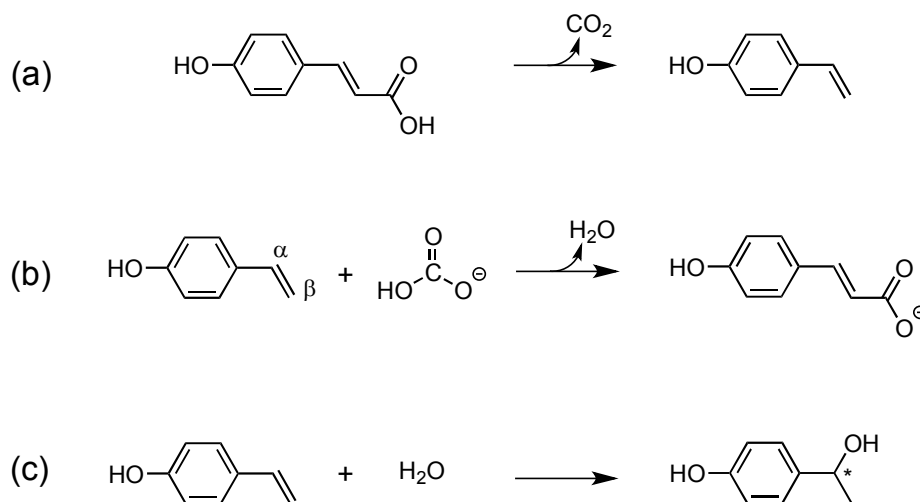


Figure 1. A) Cluster model employed to investigate the peptidyl-tRNA hydrolysis reaction. B) Mechanism suggested on the basis of the calculations. Calculated relative energies in kcal/mol are indicated.

3.2. Phenolic acid decarboxylase.

The technical developments discussed above, most importantly the increasing size of active site models due to the increased computer power, have made it possible to address new kinds of mechanistic questions. One such recent study that used a large active site model concerns the reaction mechanism of phenolic acid decarboxylase (PAD), which catalyzes the nonoxidative decarboxylation of phenolic acids to their corresponding *p*-vinyl derivatives without the use of cofactors (Scheme 1a). This enzyme is of biocatalytic interest, as the styrene derivative products can be used in, e.g., the polymer and food industries. A model consisting of >300 atoms was designed (Figure 2A) and different mechanistic scenarios were investigated.[22] It was first found that the substrate binds in a different orientation as compared to the literature proposals, namely with the *p*-hydroxyl group, rather than the carboxylate, pointing toward the Tyr11 and Tyr13 residues. These interactions are predicted to lead to the deprotonation of the hydroxyl group once the substrate binds to the active site. Model calculations showed that the pK_a of a phenol drops by 3-6 units when hydrogen-bonded to two water molecules, which agrees well with measurements on the related enzyme vanillyl-alcohol oxidase.[29]



Scheme 1. Reactions investigated with the PAD active site model. a) The natural decarboxylation of phenolic acids, b) The promiscuous carboxylation of hydroxystyrenes, and c) the enantioselective hydration of hydroxystyrenes.

Next, the calculations showed that the Glu64 residue acts as a general acid, protonating the double bond of the substrate to give the postulated quinone methide intermediate (Figure 2B). The overall energy barrier for this step was calculated to be 16.0 kcal/mol. Previously, Tyr19 was proposed to be the general acid in the reaction, but the calculations showed that it is not acidic enough to play this role. The final step is the C-C bond cleavage to generate the vinyl phenol product and CO₂, and the barrier for this step was calculated to be very similar to the first step (15.9 kcal/mol). In calculating the energy associated with the release of the CO₂ gas, the translational entropy contribution was included, according to the protocol described above. The mechanism proposed on the basis of the calculations is consistent with mutagenesis experiments, and the calculated energy barrier is in agreement with measured rate constants.[23]

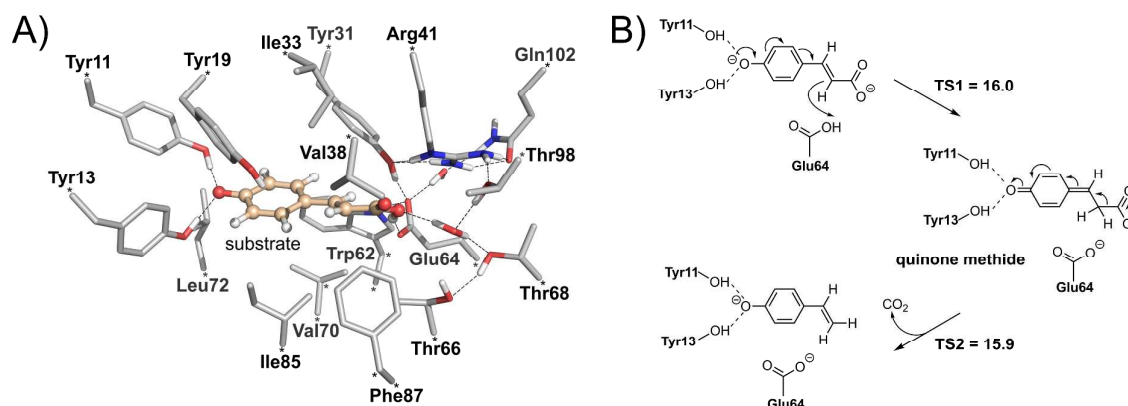


Figure 2. A) Model employed to investigate the mechanism of phenolic acid decarboxylase (PAD) B) Mechanism suggested on the basis of the calculations. Barriers in kcal/mol are indicated.

In a subsequent study, the same active site model was used to investigate two promiscuous activities of PAD, namely the carboxylation and hydration of hydroxystyrenes (Scheme 1b,c).[30] On the basis of the calculations, new mechanisms were proposed for these reactions, which differed substantially from the literature proposals.[31] In the case of carboxylation, the calculations suggested that carbon dioxide is first formed from bicarbonate by a proton transfer from Glu64. CO₂ can then form the C-C bond with the substrate, which is the reverse of the decarboxylation reaction discussed above.

For the hydration reaction, on the other hand, a quinone methide intermediate was suggested to be formed first by a proton transfer from Glu64 to the C=C double bond of the substrate, after which a water molecule, activated by the bicarbonate, performs a nucleophilic attack at the α -carbon to yield the alcohol product. Most importantly, the calculations were able to reproduce the enantioselectivity of the hydration reaction. Experimentally, the *S*-alcohol was observed with up to 87% ee,[31] which corresponds to an energy difference of 1.6 kcal/mol. The calculations showed that the transition state for forming the *R*-alcohol is higher by 2.3 kcal/mol, in good agreement with the experimental value. By analyzing the lowest-energy TS structures leading to the *S*- or *R*-products (Figure 3), a plausible explanation for the enantioselectivity could be formulated and key active site residue could be identified.[30] Here, it is important to point out that in order to obtain reliable energy differences and thus reproduce the enantioselectivity, a large number of enzyme-substrate complexes had to be considered, since the substrates of this reaction (hydroxystyrene, water and bicarbonate) can bind to the active site in many different ways.

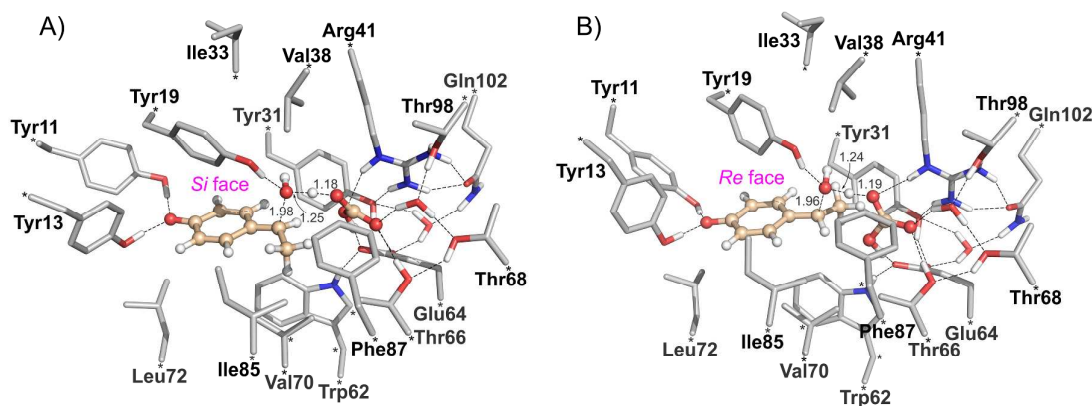


Figure 3. Optimized structures for the lowest-energy transition states leading to the *S*- and *R*-products in the hydration reaction catalyzed by phenolic acid decarboxylase.

3.3. Modeling of enantioselectivity.

The study of stereoselectivity of enzymes with large cluster models is a new exciting development in the field. The cluster approach has previously been quite successful in

reproducing and rationalizing various kinds of selectivities in enzymatic reactions, and there are numerous examples of studies in which limited active site models have been able to elucidate sources of, e.g., chemoselectivity, regioselectivity, and substituent effects.[1] This has been particularly successful for metalloenzymes, where the selectivity to a large extent is dictated by the metal site and groups in its immediate vicinity. A general challenge in studying selectivities is that small energy differences, on the order of 1-2 kcal/mol, must be calculated in order to reproduce the experimental observations. The success of the cluster approach in this context relies of course to a large extent on a cancellation of errors between very similar transition state structures, which results in a higher accuracy.

The study of enantioselectivity poses additional challenges, since the chiral active site environment around the substrate must be correctly accounted for in order to capture the effects that induce the selectivity. This requires sufficiently large active site models, something that has become possible only quite recently. We had previously, in a proof-of-concept study, examined the enantioselectivity of limonene epoxide hydrolase (LEH), an enzyme that catalyzes the hydrolysis of epoxides to their corresponding vicinal diols.[32] Since the epoxide can be opened at either of the two carbons, two chiral diol products can be formed. A model of the active site consisting of 259 atoms was designed and the regioselectivity of the ring opening of the non-natural substrate *meso*-cyclopentene was considered for both the wild-type and various engineered variants of LEH that yield high enantioselectivity for either the (*R,R*) or the (*S,S*) product.[33] The agreement with the experimental findings was very good, considering the small energy differences involved. Importantly, the calculations were also able to provide a rationalization of the sources of enantioselectivity in the various mutants. These results were very encouraging and the same methodology has subsequently been applied to several other enzymes, such as aryl malonate decarboxylase [22] and soluble epoxide hydrolase.[34] Considering the amounts of detailed information provided by this kind of active site models, it is expected that the cluster approach will become a very valuable tool in the field of asymmetric biocatalysis.

3.4. *Combination with other methodologies.*

Another fruitful recent application of the cluster approach is to generate the potential energy profile for an enzymatic reaction, which is then used for the parameterization of empirical valence bond (EVB) Hamiltonian. The EVB simulations can in turn be used to ask other questions about enzyme catalysis. A very interesting example concerns the role of entropy in enzyme catalysis and the so-called Circe effect, which hypothesizes that part of the free energy of the substrate binding is used to reduce the entropic penalty of the chemical step of the enzyme.[35] The specific enzyme for which this hypothesis was examined is cytidine deaminase (CDA), a zinc-dependent enzyme that catalyzes the hydrolytic deamination of cytidine to uracil. Measurements have namely shown that substrate binding in CDA is associated with an entropy loss of similar magnitude as the activation entropy penalty for the uncatalyzed reaction in water, and also that the activation entropy for the chemical step is almost zero.[36]

First, a cluster model of the active site was designed, consisting of 191 atoms (Figure 4), with which the detailed catalytic cycle was established and the potential energy graph was obtained.[20] These energies were then used to construct EVB models for the various steps of the reaction, which in turn were used to perform extensive sampling of the reaction coordinate using molecular dynamics and calculate free energy profiles at different temperatures. The obtained Arrhenius and van't Hoff plots could then be used to calculate the activation and reaction entropies and enthalpies of the individual steps, which could be compared directly to the measured values for both the enzyme and the uncatalyzed water reaction. The agreement with the experimental results was very good. In particular, the calculations could reproduce the very small entropy contribution measured for the rate-limiting step, but the reason for the drop in entropic penalty was found to be a different one compared to the Circe hypothesis. Namely, the calculations showed that the reaction mechanism changes between the uncatalyzed solution reaction and the enzymatic one. The enzymatic reaction proceeds through a stepwise hydroxide attack mechanism, while the uncatalyzed reaction proceeds through a concerted mechanism. These results therefore speak against the Circe hypothesis. Rather, the calculations showed that the active site of the enzyme is preorganized to stabilize a reaction mechanism that is different

from the uncatalyzed one taking place in water. Evidently, this combined cluster/EVB approach is very insightful and we expect therefore that this kind of applications will be more common in the future.

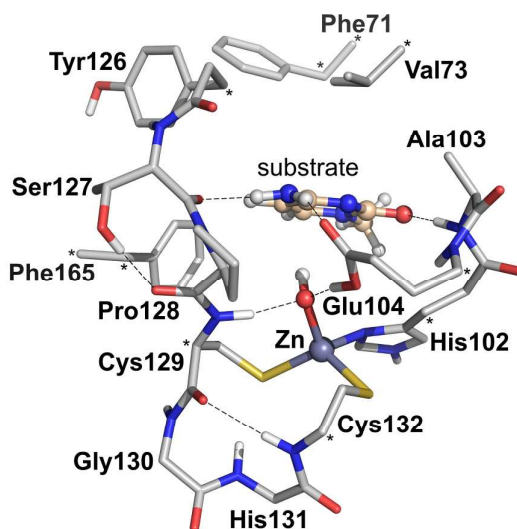


Figure 4. Active site model of cytidine deaminase used to investigate the Circe effect.

Another interesting example of the application of the cluster approach for parameterization and virtual screening is worth mentioning here. Sköld and co-workers employed a cluster model of the active site of insulin-regulated aminopeptidase (IRAP) consisting of 227 atoms to optimize the geometries of the intermediates and transition states along the reaction pathway of this zinc-dependent enzyme.[37] The geometries were then used to conduct a docking and virtual screening study to find potential inhibitors. The developed protocol could identify a number of known inhibitors from a large library of compounds, and, very importantly, also identify a new compound not previously recognized as inhibitor for this enzyme.[37] Also here, considering the successful outcome of this study, it is expected that this kind of applications will increase in the future.

As a final example of different applications of the cluster approach we mention here the work of Jensen and co-workers.[38] The results of cluster models of five different enzymes calculated with the B3LYP functional were used to assess a number of semi-empirical methods in terms of barrier heights and reaction energies. This work represents a first step in the

creating of a benchmark set for barriers calculated for large systems relevant for enzymatic reactions, something that can become very valuable in order to test and parametrize new methodologies that can be used in, e.g., more accurate fast virtual screening for drug discovery purposes.

4. Concluding Remarks

To summarize, we have in this Perspective discussed the latest methodological developments in the cluster approach for studying enzymatic reactions and given examples of some recent applications. It has in the last fifteen years or so been demonstrated that this methodology is extremely valuable for elucidating reaction mechanisms. Different mechanistic scenarios can be examined relatively fast by comparing their energy profiles, something that has been applied to a large number of diverse enzymatic systems.[1]

Over the years, the methodology has been gradually refined, as practical experience is gained and the techniques are developed. The exponential growth in computer power has generally translated into larger models of the active sites, which has allowed for addressing more complicated problems. The study of stereoselectivity in enzymes is an example of such questions that were not tractable a few years ago. However, even if it becomes possible to treat much larger models, by, for example, linear scaling methods,[39] there is a limit before the size starts to cause problems in terms of multiple minima, and sampling becomes necessary. Already today, with models of ca 300 atoms, one has to be very careful in order to make sure that no artificial movements take place between the different stationary points, something that can have severe consequences and can lead to wrong conclusions. This is particularly important in the study of selectivity, which requires reproducing very small energy differences. Therefore, the future enhancement in computer power is likely to be invested in more accurate electronic structure methods to improve accuracy. Examples of such development have already started to appear.

In the coming years, the cluster approach will undoubtedly continue to have a strong impact on the field of mechanistic enzymology. The constant improvement of the methodology coupled with the continuous increase in computer power, will allow new kinds of applications

and lead to breakthroughs both in the fundamental understanding of enzymatic reactions, and also in the utilization of these in industrial processes.

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Table of Contents Graphics

